Last Time

- Independent events A & B:

$$P(A \cap B) = P(A)\,P(B)$$

- Mutually exclusive events A_1 & B_1:

$$A_1 \cap B_1 = \emptyset$$

Three independent events

Three events A_1, A_2, and A_3 are said to be mutually independent if

(i) A_i and A_j are independent, $\forall\, i \neq j$.

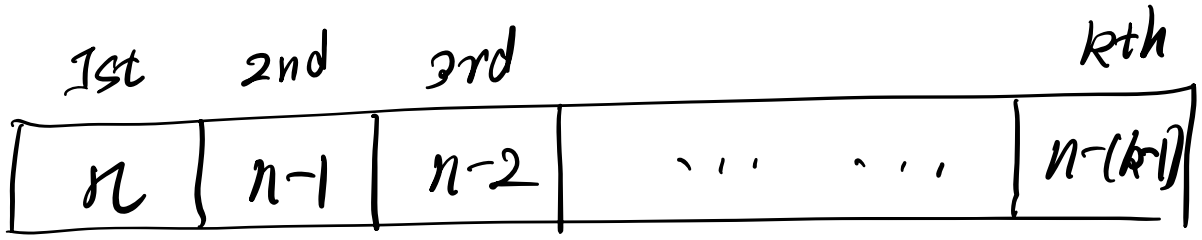
(ii) $P(A_1 \cap A_2 \cap A_3) = P(A_1)\,P(A_2)\,P(A_3)$

Generalization

An experiment consists of two subexperiments. If one subexperiment has a outcomes and the other subexperiment has b outcomes, then the experiment has ab outcomes.

(k subexperiments)

Example We choose an ordered sequence of k distinguishable objects out of a set of n distinguishable objects. An ordered sequence of k distinguishable objects is called a **k-permutation**. What is the number of k-permutations of these n distinguishable objects?



Subexperiment 1

The number of k-permutations of n distinguishable objects is

$$(n)_k = n\cdot(n-1)(n-2)\cdots\cdots(n-(k-1))\cdot\frac{(n-k)\cdots 3\cdot 2\cdot 1}{(n-k)\cdots 3\cdot 2\cdot 1}$$

$$= \frac{n!}{(n-k)!}$$

Sampling without Replacement

Example. Suppose we have four distinct objects,

$$\mathcal{S} = \{A, B, C, D\}.$$

We select two objects **sequentially**. After the first object is selected, it is *not returned* to the collection before the second selection.

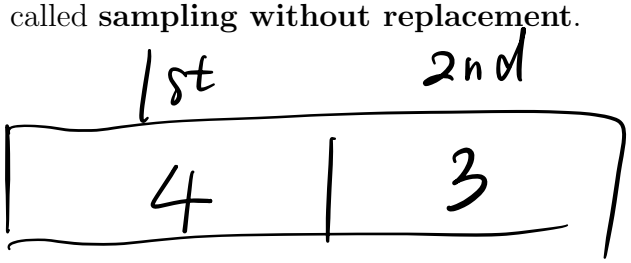
For example, if the first selected object is A, then the second object must be selected from

$$\{B, C, D\}.$$

Thus, the set of available objects changes after the first selection:

$$\{A, B, C, D\} \xrightarrow{\text{select } A} \{B, C, D\}.$$

This procedure is called **sampling without replacement**.



$4 \cdot 3 = 12$ outcomes

Suppose the **order of selection is recorded**. Then, the sample space is

$\Omega_{\text{order}} =$ { AB, AC, AD,
BA, BC, BD,
CA, CB, CD,
DA, DB, DC }

Now suppose that after selecting the two objects, we arrange them **alphabetically** and record only the resulting pair. Thus, the order in which the objects were selected is not recorded.

For example,

$$AD \quad \text{and} \quad DA$$

both produce the recorded outcome

$$AD.$$

Therefore, the sample space of the recorded experiment is

$$\Omega = \{AB, AC, AD, BC, BD, CD\}$$

Hence, the number of possible outcomes is 6 and denoted as follows:

$$|\Omega| = \binom{4}{2} = 6.$$

"4 choose 2"

Key Idea: In sampling without replacement, a selected object is unavailable for subsequent selections. Consequently, the successive selections are generally dependent.

The general case

Suppose we have n distinguishable objects and want to choose k of them, without regard to order. Each such selection is called a **k-combination**.

We use the notation

$$\binom{n}{k}$$

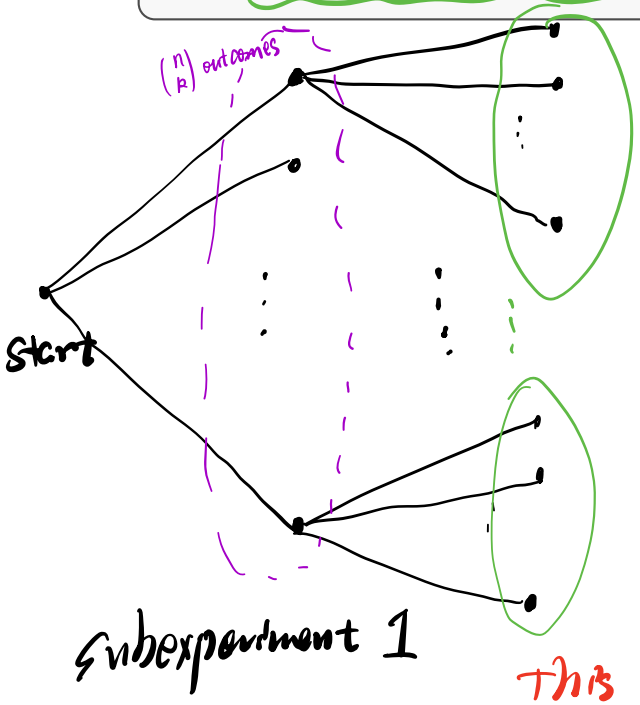
to denote the number of k-combinations of n objects, read as "n choose k."

An Experiment

To determine $\binom{n}{k}$, consider the following two **subexperiments** for constructing an ordered selection of k objects from n objects:

1. Choose a k-combination from the n objects.
2. Arrange the k selected objects in an order.

$(n)_k = n\cdot(n-1)\cdots(n-(k-1))$



$k\cdot(k-1)\cdots 2\cdot 1 = k!$

k	$k-1$	$\cdots$	1

$$\binom{n}{k}\cdot k! = (n)_k = \frac{n!}{(n-k)!}$$

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

This is also called binomial coefficient as it is from a binomial: $(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}\cdot b^k$

Interpretation of $\binom{n}{k}$

$\binom{n}{k}$ is the number of ways to select k objects from n distinct objects when

- sampling is without replacement, and
- the order of selection does not matter.

Example. A standard deck contains 52 cards, of which 4 are queens. Suppose 7 cards are selected uniformly at random without replacement, and the order of selection is not recorded. What is the probability that you have no queens?

An outcome of this experiment is a 7-combination of the 52 cards

The sample space Ω has $\binom{52}{7}$ outcomes.

We want to know the number of 7-combinations that have no queens:

$\binom{48}{7}$ select 7-combinations from $52-4=48$ non-queen cards.

Let A denotes the event that the selected 7 cards contains no queens.

Then, $|A| = \binom{48}{7}$

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\binom{48}{7}}{\binom{52}{7}} \approx 0.55036$$



Sampling with Replacement

$n \cdot n \cdot n \cdots n$ (m times)

Given n distinguishable objects, there are n^m ways to choose with replacement an *ordered* sample of m objects.

Example. There are ten students in a probability class. Each student receives one of four possible grades:

$$\mathcal{S} = \{A, B, C, F\}.$$

We record the grades of the students sequentially. Let x_i denote the grade of the ith student. For example, one possible outcome is

$$x_1 x_2 \cdots x_{10} = CBBACFBACF.$$

For each student, there are 4 possible grades. The same grade can occur for more than one student, so this can be viewed as **sampling with replacement** from the set

$$\{A, B, C, F\}.$$

Therefore,

$$|\Omega| = \underbrace{4 \times 4 \times \cdots \times 4}_{10 \text{ times}} = 4^{10} = 1,048,576.$$

Summary of Sampling Methods

Suppose we select k objects from a set of n distinguishable objects.

Sampling Method	Number of Outcomes
With replacement, order matters	n^k
Without replacement, order matters	$(n)_k = \dfrac{n!}{(n-k)!}$ $= n \cdot (n-1) \cdot (n-2) \cdots (n-(k-1))$
Without replacement, order does not matter	$\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$